

Trigonometric functions and equations - Set 1

TMUA practice questions by topic. Specification MM4.3-MM4.6. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

- A 2
- B 6
- C 4
- D 8
- E 1
- F 3

2 Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

- A $\frac{\sqrt{3}}{3^2}$
- B $\frac{4\sqrt{3}}{4}$
- C $\frac{\sqrt{3}}{4}$
- D 1
- E 0
- F $\frac{1}{2}$

3 Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

- A $\frac{2\pi}{3}$
- B $\frac{4\pi}{3}$
- C $\frac{\pi}{3}$
- D $\frac{5\pi}{3}$
- E $\frac{2\pi}{3}$
- F $\frac{7\pi}{3}$

- 4 Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.
- A $-\frac{3}{4}$
 B $\frac{3}{4}$
 C $-\frac{3}{5}$
 D $-\frac{3}{5}$
 E $\frac{4}{5}$
 F $-\frac{4}{5}$
- 5 Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?
- A 3
 B 6
 C 2
 D 8
 E 4
 F 1
- 6 Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.
- A π
 B $\frac{5\pi}{6}$
 C $\frac{7\pi}{6}$
 D $\frac{\pi}{6}$
 E $\frac{3\pi}{2}$
 F $\frac{2\pi}{3}$
- 7 Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.
- A 1
 B $\sin \theta$
 C $\tan \theta$
 D $\sin^2 \theta$
 E $\cos \theta$
 F $\frac{\sin \theta}{\cos \theta}$

- 8 The function $f(x) = 3\sin(2x) - 1$ is defined for all real x . Find the difference between its maximum and minimum values.
- A 1
B 2
C 4
D 3
E 6
F 7
- 9 Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.
- A 1
B 4
C 6
D 3
E 8
F 2
- 10 For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals
- A $2\sin^2 \theta$
B $2 + 2\sin \theta \cos \theta$
C 0
D 2
E $2\cos^2 \theta$
F 1
- 11 Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .
- A π
B $\frac{5\pi}{2}$
C $\frac{7\pi}{2}$
D $\frac{4}{3}\pi$
E $\frac{3\pi}{2}$
F 2π

12 Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

A $\frac{\sqrt{2}}{2}$

B 1

C $\frac{3}{2}$

D 2

E $1 + \sqrt{3}$

F $\frac{1}{2}$